HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented query optimization strategies for a distributed document database, improving query execution times by an average of **28%** for analytical workloads.
- Developed new features for the query engine's parsing and optimization phases, contributing to the next major version of the database system.
- Refactored and enhanced the geospatial search subsystem, enabling more complex and performant location-based queries for enterprise customers.
- Integrated diagnostic tools for query performance problems into the core engine, aiding support and consulting teams in troubleshooting customer issues.
- Collaborated with client drivers and cloud services teams to ensure seamless integration of query engine enhancements across different platforms.
- Contributed to code reviews and design discussions for new query performance features, adhering to strict code quality and architectural standards.
- Containerized query engine components using **Docker** for local development and testing environments, streamlining the build and deployment pipeline.

SKILLS

Core Engineering – Data Structures, Algorithms, Software Design, Software Architecture, Distributed Systems, Database Systems, Query Optimization, Query Language

Languages & Databases – SQL, Java, Python, C++, MongoDB

Tools & Technologies – Jira, Confluence, Git, Jenkins, Docker, Linux

Cloud & Platforms – AWS, CloudWatch

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, R, pandas, NumPy, scikit-learn

PROJECTS

Distributed Query Engine Optimization (HCL Technologies)

- Enhanced the query optimizer in a large-scale enterprise database system, reducing average query latency by **35%** for common analytical patterns.
- Implemented parallel execution strategies for complex SQL queries across distributed nodes, improving throughput by approximately **22%**.
- Developed unit and integration tests using **JUnit** and **Mockito** to ensure the robustness and correctness of query plan generation logic.

Geospatial Indexing for Transactional Data

- Designed and built new indexing structures to accelerate geospatial queries on a large transactional dataset, cutting query times for location-based searches by over **40%**.
- Integrated the new geospatial index with the existing query execution engine, ensuring compatibility and minimal impact on other query types.
- Wrote comprehensive documentation and provided internal training on the usage and performance characteristics of the enhanced geospatial features.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)